Meeting Notes — Dr. Moon

Date: May 22, 2026 **Attendees:** Aiwei, Nimra, Dr. Moon **Context:** Walkthrough of revised pipeline (LLM1 extraction → LLM2 ISO time-tagging → LLM3 dedup/combine) and clinical-grounding check on the timeline output. Paper target: ACL via ARR, deadline ~Aug 3, 2026 (74 days).

TL;DR

Dr. Moon validated the **machine-readable / intermediate-representation framing** as the right direction, but was clear that a clinician would prefer the original discharge summary narrative for direct reading. The most clinically compelling use case she surfaced — unprompted — was **cross-document discrepancy detection**: using timelines from multiple notes (e.g., discharge summary + follow-up PCP note) to flag inconsistencies, typos, medication errors, and gaps. She also pushed on formatting refinements (concision, SOAP-style grouping, list/table format for labs) that would make the timeline more readable if a clinical user study were ever attempted later.

Dr. Moon's Position on the Current Output

- **The intermediate-representation framing is solid.** Reconciling discharge summaries across sites, providers, and formats is a real problem, and machine-readable timelines make sense as a substrate for that. She explicitly said the rationale "makes a lot of sense to me."
- **As a direct clinical reader, she would still prefer the left-hand-side narrative.** Trained clinicians are habituated to the H&P/SOAP narrative format. The right-hand timeline reads as "clunky" because of repetition (e.g., "Upon admission the patient was…" stacked five times) and atomized phrasing of physical exam and lab data.
- **She does not see a current clinical workflow where she would open the timeline instead of the discharge summary.** This is honest feedback we should take seriously.
- **She was not categorical that this is useless** — she was specific that the value lies elsewhere than direct narrative replacement.

The Discrepancy-Detection Idea (Dr. Moon's Contribution)

This is the strongest clinical framing that came out of the meeting and we should anchor on it.

- Clinical documentation quality varies widely. Providers make typos, write circular logic ("hypovolemia due to hypovolemia"), confuse medications, write fragmentary hospital courses, and copy-forward stale information.
- When multiple notes describe the same encounter or trajectory (e.g., a discharge summary + a PCP follow-up note 2 days later), discrepancies between them are clinically meaningful and currently invisible to any automated system.
- A structured, time-grounded timeline per note enables **cross-timeline comparison**: pointing out where Note A and Note B make assertions that cannot both be true, without claiming ground truth ("reporting the news" framing — A says X, B says Y).
- **Especially high-value at points of handoff** — night-to-day nursing, inpatient-to-outpatient, ED-to-floor. Handoffs are exactly where documentation errors become unsafe.
- Limitation Aiwei flagged: MIMIC-III is single-encounter per patient (mostly), so this use case is hard to demonstrate without the state-agency dataset. Likely a follow-up paper.

Format Refinements She Suggested

Ordered roughly by what's easy vs. hard to implement.

Easy / low-risk

1. **Collapse repetitive prefixes.** Don't say "Upon admission the patient had…" five times. Match the source phrasing — if the discharge summary writes "Cardiac: regular rate and rhythm," just put that.
2. **Labs as a list, not narrative.** Bullet points (more bullets, less prose per bullet) for lab values. Tabular ideal but list acceptable.
3. **Front-load admission medications / pre-existing context.** Pre-hospital meds belong near the top of the timeline because they describe "who this patient is" before the encounter. This may require an additional LLM pass.
4. **Flag abnormal findings.** If a lab or exam finding is abnormal, make it visually stand out. Not a structural change — a presentation layer.

Medium difficulty

5. **Shorter narrative-style events for plan/course statements.** Where the source already combines clauses ("seen in ED and admitted to trauma service with closed left femur fracture"), the timeline should too — splitting into atoms reads clunky. Only combine where the *source* combined; do not let the LLM invent combinations across sentences (hallucination risk).
6. **Within-day grouping by SOAP category.** Group same-date events by Subjective / Objective / Assessment / Plan. She thought this was useful for *single notes* but less so for

discharge summaries (which are inherently a summary across the whole stay). For a discharge summary specifically, the timeline structure may be more natural than SOAP.

Hard / probably out of scope for this paper

7. **Per-day high-level summary bullets.** Compressing all events on a date into a single summary sentence. Dr. Moon liked the idea; Aiwei flagged that this introduces too much LLM freedom and will be hard to evaluate. Defer.

Honest Trade-Off Dr. Moon Surfaced

> "As a human, I'd rather read the left-hand side, 100%."

She was direct that the original narrative is what a clinician would actually use. The timeline is not going to displace the discharge summary for direct clinical reading. The value is:

- Machine consumption (cohort search, temporal QA, modeling, dashboards)
- Cross-document reconciliation (discrepancy detection)
- Possibly: very condensed summary outputs *derived from* the timeline (2-sentence "what happened" for downstream consumers)

On the Date-Tagging Honesty Question

Aiwei asked about the shift from "guess a date for everything" to "[NO DATE PROVIDED] for events with no resolvable date." Dr. Moon's response: **precision-over-recall is the right call from a clinician's standpoint.** Don't fabricate. Atemporal facts (PMH, allergies, family history) don't need a date and the narrative format presenting those is already fine on the left-hand side.

Evaluation Plan — Status

- ~200 stratified samples
- Aiwei + Nimra score independently using rubric (completeness, hallucination/correctness, temporal order)
- Tuned LLM judge produces a third score
- Disagreements → error analysis
- **Open ask to Dr. Moon:** can a clinician (her, or a medical student / apprentice she can recruit) spot-review the human annotations rather than score from scratch? Dr. Moon said she cannot commit personally but will try to find someone.

Action Items

#	Owner	Item
1	Aiwei	Write up these notes + decision proposal, share with Nimra
2	Aiwei	Discuss with Mumu (currently traveling) — particularly the question of whether to fall back from "shorten more aggressively" framing
3	Aiwei	Discuss with Mohit (in Seattle, internship)
4	Aiwei	Decide on framing: (a) NLP pipeline + intermediate representation, with discrepancy detection as motivated future work, vs. (b) pivot to a more clinically-grounded HCI venue (would require user study and a different deadline)
5	Aiwei	Implement easy format refinements (collapse repetitive prefixes, list-format labs, move admission meds to top) — additional LLM call cost acceptable
6	Aiwei → Dr. Moon	Email short sample outputs as the pipeline evolves; biweekly meetings starting June 4
7	Dr. Moon	Attempt to identify a med student / apprentice who can spot-review annotations (no commitment)

Open Questions for Mumu

1. Do we agree to frame the paper around the **intermediate-representation + downstream-applications** thesis, with discrepancy detection explicitly called out as motivated future work?
2. How aggressive should we be on within-timeline summarization for this submission? Mumu was pushing for more compression; Dr. Moon's feedback supports *some* compression (collapsed prefixes, list labs) but not aggressive per-day summarization for evaluation reasons.
3. If we cannot get the state-agency data in time, are we comfortable that the discrepancy-detection follow-up is positioned as future work rather than executed in this paper?